SGA 3, Exposé XXV
The Existence Theorem
English Loop-1 Reconstruction

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Exposé XXV
The Existence Theorem

by M. Demazure

[1] For completeness, in this Exposé we give a proof of the existence theorem for split groups. Like Chevalley's original proof ("Sur certains schémas de groupes semi-simples", *Séminaire Bourbaki*, May 1961, no. 219), it rests on the existence of complex semisimple algebraic groups of every possible type. Cartier gave the principle of a more satisfactory proof, establishing directly the existence of a simply connected split semisimple $\mathbb{Z}$-group associated with a given Cartan matrix (unpublished).[2] Let us point out, however, that the difficulty is not to give an explicit construction of a group scheme, but to verify that the group so constructed really satisfies the required conditions—essentially, that its fibers are indeed smooth and reductive.

1. Statement of the theorem

Theorem 1.1.— *Let S be a nonempty scheme. The functor*

$$G \longmapsto R(G)$$

is an equivalence from the category of pinned reductive S-groups to the category of pinned reduced root data.

By the uniqueness theorem (Exposé XXIII, 4.1), the preceding statement is equivalent to the following.

Corollary 1.2 (Existence of split groups).— *For every reduced root datum R, there exists a split reductive $\mathbb{Z}$-group G such that $R(G) \simeq R$.*[3]

In particular:

Corollary 1.3.— *Let k be a field. For every splittable reductive k-group G_k, there exists a reductive $\mathbb{Z}$-group G such that*

$$G \otimes_{\mathbb{Z}} k \simeq G_k.$$

[1] Editors' note: Version of 13 October 2024.

[2] Editors' note: Such a principle was also sketched by B. Kostant *[Ko66]*; a complete proof, using quantum groups, was given more recently by G. Lusztig *[Lu09]*.

[3] Editors' note: The work *[BT84]* of F. Bruhat and J. Tits also contains a variant of Chevalley's construction (*loc. cit.*, 2.2.3–2.2.5 and §3.2), which in particular gives a smooth affine $\mathbb{Z}$-group G with connected fibers, possessing a split maximal torus, and whose generic fiber is a reductive $\mathbb{Q}$-group of type R. The reductivity of the geometric fibers of G follows from the study of the unipotent radical of a special fiber in *loc. cit.*, 4.6.12 and 4.6.15 (valid for more general G, associated with a valued root datum); but it is simpler to deduce it from the description of $\mathfrak{g} = \mathrm{Lie}(G)$, from Exposé XIX, 1.12(iii), and from the existence of the elements $w_\alpha(X)$ of Exposé XX, 3.1(iv); see *[BT84]*, 3.2.1.

Conversely, first observe that to prove 1.2 it is enough, by Exposé XXII, 4.3.1, and Exposé XXI, 6.5.10, to consider the case in which the root datum R is simply connected (and even irreducible, if desired, by Exposé XXI, 7.1.6).

On the other hand, under the hypotheses of 1.3 the group scheme G has constant type (since $\mathrm{Spec}(\mathbb{Z})$ is connected), hence type $R(G_k)$. By Exposé XXIII, 5.9, the validity of 1.3 for a given group G_k therefore entails the existence of a split $\mathbb{Z}$-group of type $R(G_k)$.[4]

Thus, to prove 1.2 and hence 1.1, it is enough to prove 1.3 when k has characteristic zero (for example, $k = \mathbb{C}$) and G_k is simply connected (and in particular semisimple), together with the following proposition.

Proposition 1.4.— *For every simply connected reduced root datum R, there exists a semisimple algebraic $\mathbb{C}$-group of type R.*

One may prove 1.4 as follows. First, there exists a complex semisimple Lie algebra $\mathfrak{g}$ of type R; see, for example, N. Jacobson, *Lie Algebras*, Chapter VII, Theorem 5.[5] Then $G = \mathrm{Aut}(\mathfrak{g})^0$ is a semisimple algebraic $\mathbb{C}$-group of type $\mathrm{ad}(R)$.[6] By *Bible*, §23.1, Proposition 1, it follows that there exists a semisimple $\mathbb{C}$-group of type R.

The remainder of this Exposé is devoted to the proof of 1.3 when k has characteristic zero and G_k is semisimple. The proof proceeds in two stages: the construction of a "piece of $\mathbb{Z}$-group scheme" (§2), and the study of the group obtained by applying "Weil's theorem" (§3).

To avoid confusion, in §2 we shall not use the abbreviated notation X_k for the k-scheme $X \otimes_{\mathbb{Z}} k$, where X is a $\mathbb{Z}$-scheme.

2. Existence theorem: construction of a piece of group

2.1.

Choose once and for all a splitting of G_k, denoted

$$(G_k, T_k, M, R)$$

(see Exposé XXII, 1.13), a system Δ of simple roots of R (defining the system R^+ of positive roots), and a Chevalley system $(X_{\alpha,k})_{\alpha\in R}$ of G_k (Exposé XXIII, 6.1 and 6.2) satisfying the additional condition (see Exposé XX, 2.6)

$$X_{\alpha,k}X_{-\alpha,k} = 1 \qquad (\alpha \in R).$$

[4] Editors' note: In fact, Exposé XXIII, 5.9 is unnecessary: the present Exposé constructs, for every semisimple $\mathbb{Q}$-group $G_{\mathbb{Q}}$, a semisimple $\mathbb{Z}$-group G of the same type as $G_{\mathbb{Q}}$, equipped with a split maximal torus T; hence G is split by Exposé XXII, 2.2.

[5] Editors' note: Let Δ be a base of R, and let $\widetilde{\mathfrak{g}}$ be the complex Lie algebra generated by $(e_\alpha, f_\alpha, h_\alpha)_{\alpha\in\Delta}$, subject to the relations $[h_\alpha, h_\beta] = 0$, $[e_\alpha, f_\beta] = h_\alpha$ if $\beta = \alpha$ and $= 0$ otherwise, $[h_\alpha, e_\beta] = (\alpha^*, \beta)e_\beta$, and $[h_\alpha, f_\beta] = -(\alpha^*, \beta)f_\beta$. In *loc. cit.*, $\mathfrak{g}$ is defined as the quotient of $\widetilde{\mathfrak{g}}$ by the intersection of the kernels of its finite-dimensional irreducible representations. In *[Se66]*, §VI.5, Theorem 9 (see also *[BLie]*, VIII, §4.3, Theorem 1), it is shown that $\mathfrak{g}$ is the quotient of $\widetilde{\mathfrak{g}}$ by the relations

$$\mathrm{ad}(e_\alpha)^{1-(\alpha^*,\beta)}(e_\beta) = 0, \qquad \mathrm{ad}(f_\alpha)^{1-(\alpha^*,\beta)}(f_\beta) = 0.$$

For an explicit description of the structure constants, and in particular the choice of signs, see Exposé XXIII, 6.5 and 6.7, as well as *[Ti66]*, §4, Theorem 1, and, for types A, D, E, *[Sp98]*, 10.2.5.

[6] Editors' note: One may suppose R irreducible, hence $\mathfrak{g}$ simple. By a general argument, $\mathrm{Lie}(G)$ is the complex Lie algebra of derivations of $\mathfrak{g}$; see *[DG70]*, §II.4, Proposition 2.3. These derivations are all inner; see *[BLie]*, I, §6.1, Corollary 3 to Proposition 1. Thus $\mathrm{Lie}(G) = \mathfrak{g}$, so G has no invariant subgroup of positive dimension and is semisimple. Its root system is consequently the same as that of $\mathfrak{g}$. Moreover, the center of G acts both trivially and faithfully on $\mathfrak{g}$, and is therefore trivial; hence G is adjoint.

Finally, choose on the subgroup of M generated by R a total order compatible with the group structure and such that the roots > 0 are the elements of R^+. We then write the roots as

$$-\alpha_n < -\alpha_{n-1} < \cdots < -\alpha_1 < \alpha_1 < \alpha_2 < \cdots < \alpha_n.$$

For $\alpha \in R$, let $U_{\alpha,k}$ denote the vector group corresponding to the root α, and let

$$p_{\alpha,k} \colon \mathbb{G}_{a,k} \xrightarrow{\sim} U_{\alpha,k}$$

be the isomorphism of vector groups defined by $X_{\alpha,k}$.

2.2.

The splitting of G_k includes, in particular, an isomorphism of k-groups

$$T_k \simeq D_k(M).$$

Put $T = D(M)$. This is a $\mathbb{Z}$-torus, and the preceding isomorphism may be regarded as an isomorphism

$$T_k \simeq T \otimes_{\mathbb{Z}} k.$$

We have

$$\mathrm{Hom}_{\mathbb{Z}\text{-gr.}}(T, \mathbb{G}_m) = M,$$

and shall regard the elements of $R \subset M$ as characters of T. Likewise, the elements of R^* will be regarded as morphisms of $\mathbb{Z}$-groups $\mathbb{G}_m \to T$.

2.3.

For each $\alpha \in R^+$, let $\mathbb{G}_a(\alpha)$ be a copy of $\mathbb{G}_a$, and consider the $\mathbb{Z}$-scheme

$$U = \mathbb{G}_a(\alpha_1) \times \cdots \times \mathbb{G}_a(\alpha_n).$$

If U_k denotes the unipotent part of the Borel subgroup B_k of G_k defined by R^+, let

$$a \colon U \otimes_{\mathbb{Z}} k \xrightarrow{\sim} U_k$$

be the isomorphism of k-schemes defined by

$$a(x_1, \ldots, x_n) = p_{\alpha_1,k}(x_1) \cdots p_{\alpha_n,k}(x_n).$$

2.4.

The group law of U_k is expressed by relations of the form

$$a(x_1, \ldots, x_n) a(y_1, \ldots, y_n) = a(z_1, \ldots, z_n),$$

where, for $h = 1, \ldots, n$, each z_h is a polynomial

$$z_h = x_h + y_h + Q_h(x_1, \ldots, x_{h-1}, y_1, \ldots, y_{h-1}),$$

whose coefficients are integers (Exposé XXII, 5.5.8, and Exposé XXIII, 6.4). Moreover,

$$Q_h(x_1, \ldots, x_{h-1}, 0, \ldots, 0) = 0.$$

Equip U with the composition law defined by these formulas, which are indeed "defined over $\mathbb{Z}$". Since this law induces the group law on U_k, it is associative, and (0) is a unit element. Indeed, these two assertions are expressed by relations among the polynomials Q_h, and $\mathbb{Z} \to k$ is injective. This is a group law: if (x_i) is a section of U over an arbitrary scheme S, its inverse (y_i) is computed recursively by

$$y_i = -x_i - Q_i(x_1, \dots, x_{i-1}, y_1, \dots, y_{i-1}),$$

formulas which are again "defined over $\mathbb{Z}$".

In summary, we have constructed a group law on U for which the isomorphism a above is an isomorphism of groups.

For each $\alpha \in R^+$, consider the morphism

$$p_\alpha \colon \mathbb{G}_a \longrightarrow U$$

defined by $p_\alpha(x) = (x_i)$, where

$$x_i = \begin{cases} x, & \alpha_i = \alpha, \\ 0, & \alpha_i \neq \alpha. \end{cases}$$

It is a closed immersion and a group homomorphism; denote its image by U_α. Since

$$(x_i) = p_{\alpha_1}(x_1) \cdots p_{\alpha_n}(x_n),$$

U is identified with the product

$$U = U_{\alpha_1} U_{\alpha_2} \cdots U_{\alpha_n}.$$

2.5.

Let $T = D(M)$ act on each U_α through the character α. One verifies immediately that this defines an action of T on the group U, and one may form the semidirect product $B = T \ltimes U$. There is a canonical isomorphism of k-groups

$$B \otimes_{\mathbb{Z}} k \xrightarrow{\sim} B_k.$$

If an arbitrary order is now placed on R^+, the morphism

$$\prod_{\alpha \in R^+} U_\alpha \longrightarrow U$$

defined by multiplication in U is still an isomorphism. Indeed, both sides are flat $\mathbb{Z}$-schemes of finite presentation, so it is enough to verify the assertion on geometric fibers. We are then reduced to Lazard's theory (*Bible*, §13.1, Proposition 1): regard U as a group with operators T, and use the fact that the U_α are pairwise nonisomorphic as groups with operators, since the characters $\alpha \in R^+$ of T are pairwise distinct on every fiber.

2.6.

Replacing R^+ by $R^- = -R^+$, one similarly constructs the groups U^-, B^-, and U_α for $\alpha \in R^-$, together with isomorphisms

$$p_\alpha \colon \mathbb{G}_a \xrightarrow{\sim} U_\alpha, \qquad \alpha \in R^-.$$

Finally introduce the product scheme

$$\Omega = U^- \times T \times U.$$

There is a canonical isomorphism of k-schemes

$$\Omega \otimes_{\mathbb{Z}} k \xrightarrow{\sim} U_k^- \times_k T_k \times_k U_k \simeq \Omega_k,$$

where Ω_k is the "big cell" of G_k (Exposé XXII, 4.1.2).

From now on we identify $\Omega \otimes_{\mathbb{Z}} k$ with Ω_k by this isomorphism, and regard U^-, T, and U as subschemes of Ω through their unit sections. Write

$$e = ((0), e, (0))$$

for the "unit section" of Ω. Our aim is now to put a piece-of-group law on Ω.

Lemma 2.7.— *Let $\alpha \in \Delta$, and let $w_{\alpha,k}$ be the element of* $\mathrm{Norm}_{G_k}(T_k)(k)$ *defined by $X_{\alpha,k}$; recall that, by definition,*

$$w_{\alpha,k} = p_{-\alpha,k}(-1)p_{\alpha,k}(1)p_{-\alpha,k}(-1).$$

There exist an open subset V_α of Ω, containing the section e, and a morphism

$$h_\alpha \colon V_\alpha \longrightarrow \Omega$$

such that:

(i) $h_\alpha(e) = e$*;*

(ii) $(h_\alpha) \otimes_{\mathbb{Z}} k$ *coincides with the restriction of* $\mathrm{int}(w_{\alpha,k})$ *to* $V_\alpha \otimes_{\mathbb{Z}} k \subset G_k$*;*

(iii) $T \subset V_\alpha$*, and* h_α *sends* T *into* T*. For every* $\beta \in R$*, one has* $U_\beta \subset V_\alpha$*, and* h_α *sends* U_β *into* $U_{s_\alpha(\beta)}$*.*

By the definition of a Chevalley system (Exposé XXIII, 6.1), for every $\beta \in R$ there exists an integer $e_\beta = \pm 1$ such that

$$\mathrm{int}(w_{\alpha,k})p_{\beta,k}(x) = p_{s_\alpha(\beta),k}(e_\beta x)$$

for every $x \in \mathbb{G}_a(S)$ and every $S \to \mathrm{Spec}(k)$.

Let S be any scheme. Write an arbitrary element of $\Omega(S)$ in the form

$$\left(\left(\prod_{\substack{\beta\in R^-\\ \beta\neq-\alpha}} p_\beta(x_\beta)\right) p_{-\alpha}(x_{-\alpha}),\ t,\ p_\alpha(x_\alpha)\left(\prod_{\substack{\beta\in R^+\\ \beta\neq\alpha}} p_\beta(x_\beta)\right)\right),$$

where arbitrary orders have been chosen on $R^- \setminus \{-\alpha\}$ and $R^+ \setminus \{\alpha\}$, as permitted by 2.5. Define a morphism $d \colon \Omega \to \mathrm{Spec}(\mathbb{Z})$ by

$$d(u) = \alpha(t) + x_\alpha x_{-\alpha}.$$

Let $V_\alpha = \Omega_d$, the open subset of Ω defined by the invertibility of $d(u)$. It contains e, T, and every U_β, $\beta \in R$. Define

$$h_\alpha \colon V_\alpha \longrightarrow \Omega, \qquad h_\alpha(u) = (a(u), b(u), c(u)),$$

where

$$a(u) = \left(\prod_{\substack{\beta \in R^- \\ \beta \neq -\alpha}} p_{s_\alpha(\beta)}(e_\beta x_\beta) \right) p_{-\alpha}\Big(-x_\alpha d(u)^{-1} \Big),$$
$$b(u) = t \cdot \alpha^*(d(u)),$$
$$c(u) = p_\alpha\Big(-x_{-\alpha} d(u)^{-1} \Big) \left(\prod_{\substack{\beta \in R^+ \\ \beta \neq \alpha}} p_{s_\alpha(\beta)}(e_\beta x_\beta) \right).$$

Since s_α permutes the positive roots distinct from α (respectively the negative roots distinct from $-\alpha$), $c(u)$ (respectively $a(u)$) is a section of U (respectively U^-), so the preceding morphism is well defined. It plainly satisfies (i) and (iii). Assertion (ii) follows immediately from the definition of the e_β, $\beta \in R$, and Exposé XX, 3.12.

Lemma 2.8.— *There exist open subsets V, V' of Ω and morphisms*

$$h \colon V \longrightarrow \Omega, \qquad h' \colon V' \longrightarrow \Omega$$

with the following properties:

(i) *V and V' contain e, and $h(e) = h'(e) = e$;*

(ii) *the morphism induced by*
$$h' \circ h \colon h^{-1}(V') \longrightarrow \Omega$$
is the restriction of the identity morphism;

(iii) *the k-morphisms $h \otimes_{\mathbb{Z}} k$ and $h' \otimes_{\mathbb{Z}} k$ are the restrictions, to $V \otimes_{\mathbb{Z}} k$ and $V' \otimes_{\mathbb{Z}} k$, of automorphisms of the group G_k;*

(iv) *V and V' contain U, T, U^-; the morphisms h, h' send U into U^-, U^- into U, and T into T.*

Let w^0 be the element of the Weyl group of G_k that sends R^+ to R^-. Write

$$w^0 = s_{\alpha_n} \cdots s_{\alpha_1}, \qquad \alpha_i \in \Delta$$

(with no relation to the numbering in 2.1), and put

$$w_0 = w_{\alpha_n,k} \cdots w_{\alpha_1,k} \in \mathrm{Norm}_{G_k}(T_k)(k).$$

Define recursively, for $i \leq n$, an open subset V_i of Ω and a morphism $h_i \colon V_i \to \Omega$ by

$$V_0 = \Omega, \qquad h_0 = \mathrm{id},$$

and, for $i = 0, \ldots, n-1$,

$$V_{i+1} = h_i^{-1}(V_{\alpha_{i+1}}), \qquad h_{i+1} = h_{\alpha_{i+1}} \circ h_i,$$

where V_{α_j} and h_{α_j} are those of 2.7.

Take $V = V_n$ and $h = h_n$. The conditions (i), (iii), and (iv) concerning V and h hold: for (i) and (iii) this follows immediately from 2.8, and for (iv) from the fact that $h \otimes_{\mathbb{Z}} k$ is the restriction of $\operatorname{int}(w_0)$ to $V \otimes_{\mathbb{Z}} k$.

Since $(w^0)^2 = 1$, we also have

$$w^0 = s_{\alpha_1} \cdots s_{\alpha_n} = (s^3_{\alpha_1}) \cdots (s^3_{\alpha_n}).$$

Put

$$w'_0 = (w_{\alpha_1,k})^3 \cdots (w_{\alpha_n,k})^3.$$

Repeating the construction above gives an open subset V' and a morphism h' that also satisfy (i), (iii), and (iv). Moreover, $h' \otimes_{\mathbb{Z}} k$ is the restriction of $\operatorname{int}(w'_0)$ to $V' \otimes_{\mathbb{Z}} k$. For every simple root $\alpha \in \Delta$, however, $(w_{\alpha,k})^4 = e$ (Exposé XX, 3.1), hence $w'_0 w_0 = e$. It follows that $h' \circ h$ induces the identity morphism on a nonempty open subset of $\Omega \otimes_{\mathbb{Z}} k$. Since Ω is smooth and of finite presentation over $\mathbb{Z}$, $\Omega \otimes_{\mathbb{Z}} k$ is schematically dense in Ω; this proves (ii).

Proposition 2.9.— *There exist an open subset V_1 of $\Omega \times \Omega$, an open subset V_2 of Ω, and morphisms*

$$\pi \colon V_1 \longrightarrow \Omega, \qquad \sigma \colon V_2 \longrightarrow \Omega$$

having the following properties:

(i) *if $x \in \Omega(S)$, then (e, x) and (x, e) are sections of V_1, and*

$$\pi(e, x) = \pi(x, e) = x;$$

(ii) *V_2 contains e, and $\sigma(e) = e$;*

(iii) *π_k and σ_k are the restrictions of the morphisms $G_k \times_k G_k \to G_k$ and $G_k \to G_k$ defined respectively by multiplication and inversion.*

Proof. Let (v, t, u) and (v', t', u') be two sections of Ω. By 2.8(iv), $h(u)$ is a section of U^- and $h(v')$ a section of U; thus $(h(u), e, h(v'))$ may be regarded as a section of Ω. Define V_1 as the open subset of $\Omega \times \Omega$ specified by

$$((v, t, u), (v', t', u')) \in V_1(S) \quad \Longleftrightarrow \quad (h(u), e, h(v')) \in V'(S),$$

using the notation of 2.8. If the pair on the left is a section of V_1, then $h'(h(u), e, h(v'))$ is defined and is a section of Ω; write

$$h'(h(u), e, h(v')) = (v'', t'', u'').$$

Set

$$\pi\big((v, t, u), (v', t', u')\big) = \big(v \cdot t v'' t^{-1},\ t t'' t',\ t'^{-1} u'' t' \cdot u'\big).$$

Assertion (i) follows immediately from 2.8(ii). To verify the assertion about π in (iii), observe that

$$h'(h(u), e, h(v')) = uv' = v'' t'' u''$$

whenever $u \in U(S)$, $v' \in U^-(S)$, and $S \to k$, by 2.8(iii) and (ii).

The morphism σ is constructed similarly. If (v, t, u) is a section of Ω, then $h(u^{-1})$ is a section of U^-, $h(v^{-1})$ a section of U, and $h(t^{-1})$ a section of T. Thus

$$(h(u^{-1}), h(t^{-1}), h(v^{-1}))$$

is a section of Ω. Define an open subset V_2 of Ω by

$$(v,t,u) \in V_2(S) \quad \Longleftrightarrow \quad (h(u^{-1}), h(t^{-1}), h(v^{-1})) \in V'(S),$$

and define

$$\sigma(v,t,u) = h'(h(u^{-1}), h(t^{-1}), h(v^{-1})).$$

The required properties of σ are verified in the same way.

Corollary 2.10.— *π is "generically associative" and σ is a "generic inverse": if $x, y, z \in \Omega(S)$, and if the expressions below are defined (as they always are over an open subset of Ω containing the unit section), then*

$$\pi(x, \pi(y,z)) = \pi(\pi(x,y), z), \qquad \pi(x, \sigma(x)) = e = \pi(\sigma(x), x).$$

Indeed, the two sides of each formula define morphisms between smooth $\mathbb{Z}$-schemes of finite presentation, and the morphisms agree on the generic fibers by 2.9(iii).[7]

Corollary 2.11.— *Let $\alpha \in R$, and let S be a scheme. Suppose that $x, y \in \mathbb{G}_a(S)$ satisfy*

$$(p_\alpha(x), p_{-\alpha}(y)) \in V_1(S), \qquad 1 + xy \in \mathbb{G}_m(S).$$

These conditions define an open subset of $\mathbb{G}_{a,S}^2$ containing the section $(0,0)$. Then

$$\pi(p_\alpha(x), p_{-\alpha}(y)) = \left(p_{-\alpha}\left(\frac{y}{1+xy}\right), \alpha^*(1+xy), p_\alpha\left(\frac{x}{1+xy}\right)\right).$$

The proof is the same as above, by Exposé XX, 2.1.

3. Existence theorem: conclusion of the proof

For convenience, make the following definition.

Definition 3.1.— *Let S be a scheme and G an S-group scheme. The group G is called* admissible *if there is an open immersion of S-schemes*

$$i \colon \Omega_S = \Omega \times S \longrightarrow G$$

such that:

(i) *the diagram*

$$\begin{tikzcd} (V_1)_S \arrow[r] \arrow[d, "\pi"] & \Omega_S \times_S \Omega_S \arrow[r, "i \times_S i"] & G \times_S G \arrow[d, "\pi_G"] \\ \Omega_S \arrow[rr, "i_S"] & & G \end{tikzcd}$$

is commutative, where π_G denotes the multiplication morphism of G;

(ii) *there is a finite set of sections $a_j \in \Omega(S)$ such that the open subsets $i(a_j)\, i(\Omega_S)$ cover G.*

[7] The French source reads "by 2.10(iii)"; the intended reference is Proposition 2.9(iii).

By "Weil's theorem" (Exposé XVIII, 3.13(iii) and (iv)), we have:

Lemma 3.2.— *If every admissible S-group is affine whenever S is étale and of finite type over $\mathbb{Z}$, then there exists an admissible affine $\mathbb{Z}$-group.*

We now have:

Lemma 3.3.— *Let S be a scheme and G an admissible S-group. Then G is smooth and of finite presentation over S, with affine, connected, semisimple fibers.*

Since Ω_S is smooth and of finite presentation over S, with connected fibers, the same is true of G by condition (ii). To prove 3.3, we may therefore suppose that S is the spectrum of a field K. Identify Ω_K with its image in G. Plainly Ω_K is the product

$$\left(\prod_{\alpha\in R^-} U_{\alpha,K}\right) T_K \left(\prod_{\alpha\in R^+} U_{\alpha,K}\right)$$

of the subgroups T_K and $U_{\alpha,K}$, $\alpha \in R$, of G. Thus the Lie algebra of G is identified with the direct sum

$$\mathrm{Lie}(T_K) \oplus \bigoplus_{\alpha\in R} \mathrm{Lie}(U_{\alpha,K}).$$

The inner automorphism defined by a section of T_K acts on $U_{\alpha,K}$, and hence on $\mathrm{Lie}(U_{\alpha,K})$, through the character

$$\alpha \in R \subset M \simeq \mathrm{Hom}_{K\text{-gr.}}(T_K, \mathbb{G}_{m,K}).$$

The preceding decomposition of $\mathrm{Lie}(G)$ is therefore exactly its decomposition under the adjoint action of T. Thus the roots of G_K with respect to T_K are the elements $\alpha \in R$.

Apply Exposé XIX, 1.13. Let T_α be the maximal torus of $\ker(\alpha) \subset T_K$, and put

$$Z_\alpha = \mathrm{Centr}_G(T_\alpha).$$

It is enough to prove that each Z_α is reductive. But

$$Z_\alpha \cap \Omega_K = \left(\prod_{\substack{\beta\in R^- \\ \beta|_{T_\alpha}=e}} U_{\beta,K}\right) T_K \left(\prod_{\substack{\beta\in R^+ \\ \beta|_{T_\alpha}=e}} U_{\beta,K}\right).$$

The roots vanishing on T_α are the rational multiples of α, hence precisely α and $-\alpha$. Consequently,

$$Z_\alpha \cap \Omega_K = U_{-\alpha,K}\, T_K\, U_{\alpha,K}.$$

By Exposé XX, 3.4, to prove Z_α reductive it is enough to show that $U_{\alpha,K}$ and $U_{-\alpha,K}$ do not commute, and this follows immediately from 2.11.

It follows from 3.2 and 3.3 that the proof will be complete once the following lemma is proved.

Lemma 3.4.— *Let S be a locally noetherian scheme of dimension at most* 1, *and let G be a smooth S-group of finite type, with affine, connected, semisimple fibers. Then G is affine (and hence semisimple).*

Nota. In Exposé XVI it was shown that 3.4 holds without any hypothesis on S, but the proof is comparatively delicate. Since only the special case 3.4 is needed here, we give a direct proof.

Consider the Lie algebra $\mathfrak{g}$ of G, a locally free $\mathcal{O}_S$-module, and the adjoint representation

$$\mathrm{Ad}\colon G \longrightarrow \mathrm{GL}_{\mathcal{O}_S}(\mathfrak{g}).$$

To prove that G is affine over S, it is enough to prove that Ad is affine. Since G is smooth with connected fibers, it is separated over S (Exposé $\mathrm{VI_B}$, 5.5), so Ad is separated. By the result proved in the Appendix (see 4.1), it suffices to show that Ad is quasi-finite. We are thus reduced to the case in which S is the spectrum of a field. In that case G is affine and therefore semisimple, and the assertion is Exposé XXII, 5.7.14.

4. Appendix

We used the following proposition in the course of the proof.

Proposition 4.1.— *Let S be a locally noetherian scheme of dimension at most 1; let G and H be two S-group schemes of finite type; and let $f\colon G \to H$ be a quasi-finite separated group homomorphism. If G is flat over S,*[8] *then f is affine.*

We shall give the proof only when G is smooth over S, which is indeed the hypothesis satisfied in the application above.

4.2.

By EGA II, 1.6.4, we may suppose that S is reduced. By the usual techniques of passage to the limit,[9] we may suppose S local. If $\dim(S) = 0$, the assertion is trivial.[10] Suppose therefore that $\dim(S) = 1$. By faithfully flat descent, we may suppose that S is complete with algebraically closed residue field. After replacing S by its normalization $\widetilde{S}$, if necessary, we may suppose S normal (EGA II, 6.7.1, and EGA 0_{IV}, 23.1.5).[11] We are thus reduced to the case in which S is the spectrum of a complete discrete valuation ring A with algebraically closed residue field.

4.3.

Let η (respectively s) be the generic (respectively closed) point of S. Consider the image $f_\eta(G_\eta)$ of G_η in H_η. It is a closed subgroup scheme of H_η. Let H' be the schematic closure of $f_\eta(G_\eta)$ in H. Since $H' \to H$ is affine (it is a closed immersion), we may replace H by H', and thus suppose that H is flat over S and that f_η is surjective. Since f_η is finite and G, H are flat over S, we have

$$\dim(G_s) = \dim(G_\eta) = \dim(H_\eta) = \dim(H_s).$$

4.4.

Let $H_s^0, \ldots, H_s^n$ be the irreducible components of H_s, where H_s^0 denotes the identity component, and let $z_0, \ldots, z_n$ be their generic points. Since each local ring $\mathcal{O}_{H,z_i}$ has dimension

[8] Editors' note: We have added the flatness hypothesis, which had been omitted.

[9] Editors' note: See EGA IV_3, 8.10.5(viii).

[10] Editors' note: Indeed, if $S = \mathrm{Spec}(k)$, with k a field, then f is the composite of the projection $p\colon G \to G/\ker(f)$ and a closed immersion i. Since $\ker(f)$ is finite over k, p is finite (Exposé $\mathrm{VI_B}$, 9.2).

[11] Editors' note: Indeed, the irreducible components $S_1, \ldots, S_r$ of S are complete integral noetherian local schemes. By a theorem of Nagata (EGA 0_{IV}, 23.1.5), the normalization $\widetilde{S}_i$ is finite over S_i, and hence $\widetilde{S}$ is finite over S. A theorem of Chevalley (EGA II, 6.7.1) then shows that if $f \times_S \widetilde{S}$ is affine, so is f.

at most 1, the morphism

$$G \times_H \mathcal{O}_{H,z_i} \longrightarrow \mathcal{O}_{H,z_i}$$

is affine, being quasi-finite and separated (see Exposé XVI, Lemma 4.2). Thus G is affine over H in a neighborhood of z_i. Let V be the largest open subset of H over which $G|_V$ is affine. Then V contains every z_i,[12] and therefore contains at least one closed point $y_i \in H^i_s$. Moreover, $\kappa(y_i) = \kappa(s)$, since $\kappa(s)$ is algebraically closed.

On the other hand, V is plainly stable under translation by any element $g \in G(S)$. We have $\dim(G_s) = \dim(H_s)$, and f_s is finite, so

$$f(s) \colon G^0_s(s) \longrightarrow H^0_s(s)$$

is surjective. Since A is complete and G is smooth over S, the canonical map $G^0(S) \to G^0_s(s)$ is surjective. Since $H^0_s(s)$ acts transitively on each $H^i_s(s)$, it follows that $V \supset H_s(s)$, and therefore, since $\kappa(s)$ is algebraically closed, $V \supset H_s$.[13]

Clearly $V \supset H_\eta$, since f_η is finite. Consequently, $V = H$. □

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[12]Editors' note: The following sentence has been added.

[13]Editors' note: When G is not assumed smooth over S, one may argue as follows. Let h_0 be a rational point of H^0_s, and let g_0 be a rational point of G^0_s such that $f(g_0) = h_0$. By Exposé $\mathrm{VI_B}$, 5.6.1, there is a commutative diagram

$$\begin{tikzcd} S'' \arrow[r, "g"] \arrow[d, "\pi"'] \arrow[dr, "\phi"] & G \arrow[d] \\ S' \arrow[r, "w"'] & S \end{tikzcd}$$

in which w is étale and surjective, π is finite and surjective, and $\phi^{-1}(s)$ consists of a single point s'' such that $g(s'') = g_0$. Then $G_{S''} \to H_{S''}$ is affine above a neighborhood of $h_0 y_i$, and the same is true of $G_{S'} \to H_{S'}$ (EGA II, 6.7.1), and hence of $G \to H$ by faithfully flat descent (EGA IV_2, 2.7.1(xiii)). Thus $h_0 y_i \in V$, from which it follows that V contains $H_s(s)$, and therefore H_s.

[14]Editors' note: Additional references cited in this Exposé.

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